INCH-POUND

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MILITARY SPECIFICATION

INITIATOR, PROPELLANT ACTUATED, JAU-2/A25 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the JAU-2/A25 Propellant Actuated Initiator Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures And Tables For Inspection By Attributes

MIL-STD-109 Quality Assurance Terms And Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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MIL-STD-453	Inspection, Radiographic
MIL-STD-1168	Ammunition Lot Numbering
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure

(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head Division (CAGE Codes 19200 and 19203)

838AS163	Test Fixture, 6.5 Cubic Inch Closed Bomb
9280894	Box, Fiberboard (Inner), Packing, Ammunition, For Initiator, (Gas-Subminiature)
9280895	Box, Packing, Ammunition, For Initiator, (Gas-Subminiature)
9280896	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Initiator, (Gas Subminiature)
DL11731494	Initiator, Propellant Actuated, JAU-2/A25 Assembly

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

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3.2 Conformance to documents. The JAU-2/A25 initiator covered by this specification shall be manufactured in accordance with DL11731494.

3.3 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Components.

3.4.1 Primary components. For the purpose of this specification, the PVU-1/A Ignition Device (Drawing 851AS110) is defined as a primary component. Only primary components from a single lot shall be used in a production lot of initiators; however, one primary component production lot may be used in more than one initiator production lot.

3.4.2 Ignition device. The ignition device shall conform to the applicable drawing and reference specification. Prior to assembly into an initiator, the ignition device shall be visually examined in accordance with 4.5.2.5 and radiographically examined for defects as specified in 4.6.6.1. The ignition device shall be manufactured within 2 years of assembly into the ignition device retainer (11738408) or recertified within 12 months of assembly into the ignition device retainer.

3.4.3 Ignition device sensitivity. The PVU-1/A Ignition Device, when assembled and crimped into an ignition device retainer (Drawing 11738408), shall fire when drop tested at an energy level of 25.48 inch ounces, maximum (see 4.5.3.6).

3.4.4 Leak test. Each retainer and primer assembly, Drawing 11738409, and the M104 propellant actuated initiator, P/N 10551446, shall withstand 14 psi internal-external pressure differential with a maximum permissible leak rate of 1×10^{-4} cubic centimeters per second of air (see 4.5.3.5).

3.5 Breakaway torque. Each threaded joint of the assembled initiator shall withstand a breakaway torque application of 400 inch-pounds without loosening or deformation (see 4.5.3.4).

3.6 Residual magnetism. The initiator shall not deflect the indicator of the compass more than five degrees in either direction (see 4.5.3.3).

3.7 Radiographic examination. Each ignition device assembly, retainer and ignition device assembly, and initiator assembly shall be radiographically examined for defects (see 4.6.6).

3.8 Functional.

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3.8.1 Ballistics. Initiators loaded with accepted pyrotechnic components (II-A pellets and PVU-1/A ignition device) and temperature conditioned shall produce pressure values specified in table I. Pressure values shall be recorded in a 6.5 cubic inch closed bomb, Drawing 838AS163.

3.8.1.1 Initiation. The initiator shall be fired by a source delivering an output pressure of 700 +25, -0 pounds per square inch gage (psig) at 70 ± 5°F.

3.8.2 Misfires. There shall be no misfires.

3.8.3 Mechanical failures. The initiator shall function without mechanical failure (see 6.5.2).

TABLE I. Pressure values.

Temperature (°F)	Pressure (psig) Minimum	Ignition delay (see 6.5.1) milliseconds, max
-60 to -70	300	10
65 to 75	300	10
195 to 205	300	10

3.9 Shear pin strength. The shear pin, after application of protective finish, shall have a single shear when a load of 41 ± 5 pounds is applied at the steady rate of 0.125 inch per minute (see 4.5.3.1).

3.10 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration (see 4.5.3.2).

3.11 Workmanship. Initiators shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.11.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.11.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.11.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550 and MIL-STD-453.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified in the contract or purchase order (see 6.2), inspection conditions shall be as follows:

a. Temperature: Room ambient 20° to 25° C (68° to 77° F)
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric: 14.7 psia

4.4 First article inspection. At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 21 assembled initiators for ballistic tests, a quantity of ignition devices in accordance with table II assembled and crimped into the retainer, five metal parts assemblies, fifty shear pins, and nine sets of unassembled parts for visual and dimensional examination and tests, and one set of unassembled components without plating or final protective finish. The sample shall be manufactured using the same materials, equipment, processes, and procedures as used in regular production. All parts and

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materials, including packaging, shall be the same as used in regular production and shall be obtained from the same source of supply.

4.4.1 Examination, tests, and facility. After provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract (see 6.2).

4.4.2 Sample size for primer sensitivity. The number of ignition devices, assembled and crimped into the retainer, submitted for ignition device sensitivity tests shall be in accordance with table II.

TABLE II. Ignition device sensitivity.

Lot Size	Sample Size
3 to 15	10
16 to 25	10
26 to 40	10
41 to 65	10
66 to 110	10
111 to 180	15
181 to 300	25
301 to 500	30
501 to 800	35
801 to 1300	40
Over 1300	50

4.4.3 Temperature conditioning for ballistic test. The sample shall be temperature conditioned and tested to determine compliance with 3.8.1. The sample for firing shall be conditioned to the temperatures specified in table III.

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TABLE III. Temperature conditioning for first article sample.

Temperature (° F)	Not vibrated	Vibrated
-60 to -70	3	4
65 to 75	3	4
195 to 205	3	4

4.4.4 First article failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in first article sample disapproval.

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105.

4.5.1.2 Lot size. The production lot size shall be as specified in the contract (see 6.2).

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed in accordance with the classification of defects. Four power (4x) magnification may be used for inspection of such defects as burr and foreign matter. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.5.2.1 Pin, Firing, Drawing 11738410, covering a detail of Drawing 11731492.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101 True position of O-ring groove with body		Gage
102 Concentricity of striker with body		Gage
103 True position of shear pin hole with body		Gage
104 Diameter of O-ring groove, min		Gage
105 Diameter of body, min	Gage	
106 Diameter of striker	Gage	
107 Length of striker, min	Gage	
108 Diameter of shear pin hole, max		Gage
109 Depth of shear pin hole, max		Gage

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110	Width of O-ring groove	Gage
111	Location of shear pin hole	Gage
112	Permissible variation of O-ring groove	Gage
113	Spherical radius of tip	Gage
114	Metal defective	Visual
115	Protective finish inadequate or defective	Visual
116	Burr or sharp corner, except where permitted	Visual
Minor		
201	Diameter of O-ring groove, max	Gage
202	Diameter of body, max	Gage
203	Diameter of shear pin hole, min	Gage
204	Depth of shear pin hole, min	Gage
205	Width of slot	Gage
206	Depth of slot	Gage
207	Total length	Gage
208	Length of body to O-ring groove	Gage
209	Location of marking	Visual
210	Marking missing, incorrect, or illegible	Visual
211	Surface finish improper, except as otherwise classified	Visual

4.5.2.2 Housing, Drawing 11731622, covering a detail of Drawing 11731492.

Critical		
1	Metal defective	Visual
2	Surface finish of cavity	Gage
Major		
101	Pitch diameter of external thread, min	Gage
102	Major diameter of external thread, min	Gage
103	Pitch diameter of internal thread, max	Gage
104	Minor diameter of internal thread, max	Gage
105	True position of pitch diameter of external thread with end flange diameter	Gage
106	Concentricity of pitch diameter of internal thread on large end with center cavity	Gage
107	True position of pitch diameter of internal thread with center cavity	Gage
108	True position of striker hole with center cavity	Gage
109	True position of shear pin hole with center cavity	Gage
110	True position of flats with internal thread	Gage
111	Perpendicularity of striker face with center cavity	Gage
112	Perpendicularity of striker face with internal thread on large end	Gage

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113	Diameter of shear pin hole	Gage	
114	Diameter of shear pin counterbore		Gage
115	Diameter of striker hole	Gage	
116	Diameter of bore, large end	Gage	
117	Depth of bore, large end	Gage	
118	Diameter of bore, small end	Gage	
119	Depth of bore, small end	Gage	
120	Depth of shear pin counterbore hole		Gage
121	Length of shear pin hole	Gage	
122	Length of internal thread on large end		Gage
123	Length of internal thread on small end, min		Gage
124	Length of internal thread on small end, max		Gage
125	Width of thread undercut	Gage	
126	Location of shear pin hole	Gage	
127	Diameter of center cavity	Gage	
128	Width of striker plate	Gage	
129	Diameter of counterbore, small end		Gage
130	Chamfer at mouth of small end		Gage
131	Thread lead angle	Gage	
132	Protective finish inadequate or defective		Visual
133	Burr or sharp corner	Visual	
134	Foreign matter	Visual	

Minor

201	Pitch diameter of external thread, max		Gage
202	Pitch diameter of internal threads, min		Gage
203	Width across flats	Gage	
204	Diameter of flange	Gage	
205	Width of flange	Gage	
206	Depth of counterbore	Gage	
207	Location of external thread from face		Gage
208	Location of shoulder from face		Gage
209	Width of shoulder	Gage	
210	Diameter of shoulder	Gage	
211	Total length	Gage	
212	Diameter of large end	Gage	
213	Angle at bottom of bore, small end		Gage
214	Surface finish improper, except as otherwise classified		Visual

4.5.2.3 Pin, Shear, Drawing 11731626, covering a detail of Drawing 11731492.

Critical

1	Metal defective	Visual

Major

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101	Diameter, min	Gage	
102	Total length, min	Gage	
103	Surface finish improper	Visual	
104	Protective finish inadequate or defective		Visual
105	Burr	Visual	
106	Foreign matter	Visual	

Minor

201	Diameter, max	Gage	
202	Total length	Gage	

4.5.2.4 Bracket, Mounting, Drawing 11731741, covering a detail of Drawing 11731494.

Critical
None defined

Major

101	Pitch diameter of internal thread, max		Gage
102	Minor diameter of internal thread, max		Gage
103	True position of mounting holes with pitch diameter of internal thread		Gage
104	Length of effective thread, min		Gage
105	Length of thread, max	Gage	
106	Metal defective	Visual	
107	Protective finish inadequate or defective		Visual
108	Burr	Visual	
109	Foreign matter	Visual	

Minor

201	Diameter of mounting holes	Gage	
202	Length across thread of bracket		Gage
203	Length across end of bracket (front)		Gage
204	Total length	Gage	
205	Location of mounting holes	Gage	
206	Location of large hole	Gage	
207	Height of shoulder	Gage	
208	Total height including boss		Gage
209	Diameter of bore	Gage	
210	Radii incorrect	Gage	
211	Surface finish improper	Visual	

4.5.2.5 Ignition Device (Percussion), Drawing 851AS110, covering a detail of Drawing 11738409.

Critical

1	Improper assembly	Visual/X-ray

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2	Any part missing	Visual/X-ray
3	Pellet broken, cracked, or dislodged	Visual
4	Metal defective	Visual
5	Sealing foil discolored	Visual
6	Mixed ignition devices	Visual

Major

101	Burr	Visual
102	Foreign matter	Visual

Minor
None defined

4.5.2.6 Ignition Device, Retainer, Drawing 11738408, covering a detail of Drawing 11738409.

Critical

1	Diameter of ignition device pocket	Gage
2	Depth of ignition device pocket	Gage
3	Metal defective	Visual

Major

101	Thickness of wall between ignition device pocket and relief cavity	Gage
102	Outside diameter	Gage
103	True position of relief cavity with ignition device cavity	Gage
104	True position of body with ignition device cavity	Gage
105	Protective finish inadequate or defective	Visual
106	Surface finish of ignition device pocket improper	Visual
107	Burr or sharp corner	Visual
108	Foreign matter	Visual

Minor

201	Diameter across chamfer	Gage
202	Diameter of relief cavity	Gage
203	Total length	Gage
204	Radius or chamfer missing or incorrect	Visual
205	Surface finish improper, except as otherwise classified	Visual

4.5.2.7 Retainer and Ignition Device Assembly, Drawing 11738409, covering a detail of Drawing 11731494.

Critical

1	Ignition device not properly seated	Gage

Major

101	Ignition device not properly sealed and crimped	Visual

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Minor
None defined

4.5.2.8 Initiator, M104 Assembly, Drawing 10551446, covering a detail of Drawing 11731494.

Critical		
1	Mixed ammunition	Visual
2	Metal defective	Visual
3	Vent hole obstructed	Visual
4	Marking missing, incorrect, or illegible	Visual
Major		
101	Thread damaged	Visual
102	Burr	Visual
103	Foreign matter	Visual
Minor		
None defined		

4.5.2.9 Firing Mechanism Assembly, Drawing 11731492, covering a detail of Drawing 11731494.

Critical		
1	Firing pin protrusion	Gage
2	Shear pin not engaged in firing pin	Gage
3	Striker hole in initiator housing obstructed	Visual
Major		
101	Burr	Visual
102	Foreign matter	Visual
Minor		
None defined		

4.5.2.10 Assembly, Drawing 11731494.

Critical		
1	Any part missing	X-ray/Visual
2	Improper assembly	X-ray/Visual
3	Total length, max	Gage
4	Marking missing, incorrect, or illegible	Visual
Major		
101	Burr	Visual
102	Foreign matter	Visual
Minor		

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201	Thread sealing compound missing	Visual
202	Protective cap or plug missing	Visual
203	Protective finish inadequate or defective	Visual

4.5.2.11 Box, Fiberboard (Inner), Ammunition, Drawing 9280894.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual
202	Box damaged	Visual
203	Sealing strip broken or missing	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Cut, damaged, or improperly sealed bag	Visual

4.5.2.12 Box, Ammunition, Packing, Drawing 9280895.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly engaged	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.13 Box, Fiberboard (Outer), Alternative, Ammunition, Packing, Drawing 9280896.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual

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203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Shear pin strength. Fifty pins from each lot of shear pins tested by the contractor shall be tested for single shear strength. Failure of any pin to comply with 3.9 shall be cause for rejection of the lot.

4.5.3.2 Protective finish. Unless otherwise specified, sampling plans and procedures in determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with 3.10.

4.5.3.3 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any assembly to comply with 3.6 shall be cause for rejection of the initiator.

4.5.3.4 Breakaway torque. Each initiator shall be subjected to the breakaway torque test. Failure of any assembly to comply with 3.5 shall be cause for rejection of the assembly.

4.5.3.5 Leak test. Each M104 Initiator and each retainer and ignition device assembly shall be subjected to a leak test. Failure of any M104 Initiator or retainer and ignition device assembly to comply with 3.4.4 shall be cause for rejection of the M104 Initiators or retainer and ignition device assembly. This test must be done by the contractor before final assembly.

4.5.3.6 Ignition device sensitivity. A sample of ignition devices, selected in accordance with table II from the initial production lot and each subsequent lot, shall be assembled and crimped into the retainer and tested for sensitivity.

4.5.3.6.1 All fire. One half the ignition device sample size, crimped into the retainer, shall be tested at the 25.48 inch-ounce energy level in accordance with 3.4.3. Any ignition device failing to comply with 3.4.3 shall be cause for rejection.

4.5.3.7 Functional tests. The following tests shall be performed by the Government.

4.5.3.7.1 Ballistics. Twenty-one units shall be tested to determine compliance with 3.8. The units shall be temperature conditioned as provided in table IV. Failure of any unit to comply with 3.8 shall be cause for rejection of the lot.

TABLE IV. Temperature conditioning for quality conformance sample.

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Temperature (° F)	Sample Size
-60 to -70	7
65 to 75	7
195 to 205	7

4.5.3.7.2 Retest. There shall be no retests.

4.5.3.7.3 Test failure. If a test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

4.6 Test methods and procedures.

4.6.1 Shear pin strength. After application of protective finish, the pin shall be subjected to the single shear strength test to determine compliance with 3.9. The test shall be performed in the fixture in conjunction with a tensile testing machine.

4.6.2 Protective finish. Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.6.3 Breakaway torque. A torque wrench, set at the torque requirement specified in 3.5, shall be used in conjunction with the fixture to determine the adequacy of the joint. Torques shall be applied in three places: (1) between the female body and male body; (2) between the male or female body and the firing mechanism assembly; and (3) between the mounting bracket and the firing mechanism assembly.

4.6.4 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end at the same height as the compass, in the north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.6.5 Leak test. The method and procedure for performing the leak test shall be as approved by the Government.

4.6.6 Radiographic examination. All ignition device assemblies, retainer and ignition device assemblies, and initiator assemblies shall be radiographically examined for defects.

4.6.6.1 Ignition device assembly, Drawing 851AS110. Prior to assembly into the cartridge, the ignition device shall be radiographically examined in accordance with MIL-STD-453. Each ignition device shall be X-rayed on boards or trays with the cup side perpendicular to the rays. The X-ray negatives shall be identified by item nomenclature and lot number. Ignition devices are

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to be left mounted to the boards or trays until after the X-rays have been reviewed and defects identified to facilitate removal of defective units. After screening, the X-rays with defectives identified shall be forwarded to the engineering agency designated by the procuring activity. Film interpretation of the radiographs shall verify the presence and correct positioning of the following:

a. Presence of ignition device charge
b. No material defects

4.6.6.2 Retainer and ignition device assembly, Drawing 11738409. All retainer and ignition device assemblies shall be radiographically examined in accordance with MIL-STD-453, with the exception that a retainer and ignition device assembly shall be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The retainer and ignition device assemblies shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All retainer and ignition device assemblies shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number (001) in each production lot. Any missing serial numbers are to be identified on the X-rays. The retainer and ignition device assemblies shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic identification shall include item nomenclature, the complete lot number assigned to that lot, and the span of serial numbers displayed. Radiographs of the entire production lot shall be submitted to the activity conducting the production lot acceptance tests. Defective units found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the production lot. Any assembling of the retainer and ignition device assembly into initiators prior to notification by the contracting agency of X-ray acceptance for the retainer and ignition device assembly shall be at the contractor's risk. Film interpretation of the radiographs shall verify the presence and correct positioning of the following:

a. Ignition device charge present and not cracked
b. Anvil present, seated, and not cocked
c. No material defects

4.6.6.3 Initiator assembly, Drawing 11731494. All assembled initiators shall be radiographically examined in accordance with MIL-STD-453, with the exception that an initiator assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The initiators shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All initiators shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive order beginning with the number 001 in each production lot. Any missing numbers are to be identified on the X-rays. The initiators shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the units displayed thereon. The radiographic plate identification shall include item nomenclature, the complete lot number, the span of serial numbers displayed, and the contract number. The radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the production lot acceptance test. Defective initiators found during X-ray review are to be marked on the X-ray prior to shipment and those defective items shall be removed from the

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production lot. Film interpretation of the radiographs shall verify the presence and correct positioning of the following parts:

a. Ignition device retainer, P/N 11738408
b. Ignition device, P/N 851AS110
c. M104 Assembly, P/N 10551446
d. Propellant, MIL-P-46994
e. Housing, P/N 11731622
f. Shear Pin, P/N 11731626
g. Firing Pin, P/N 11738410
h. O-ring, MS28775-009
i. Mounting Bracket, P/N 11731741

4.6.7 Ignition device sensitivity. Drop test a sample of ignition devices, assembled and crimped into the retainer, selected in accordance with table II. Use a steel ball 15/16 inches in diameter and weighing 1.94 ± 0.02 ounces at a drop height of 13 inches (measured from bottom of suspended ball to top of firing pin) in a primer sensitivity machine approved by the procuring agency.

4.6.8 Functional.

4.6.8.1 Ballistics. The test sample shall be rigidly supported in the fixture and fired by a source delivering an output of 700 +25, -0 psig at 70 ± 5°F. The sample shall be tested within 5 minutes after removal from the temperature conditioning chamber. When this time limit is exceeded, the sample units shall be reconditioned in accordance with 4.6.8.2. Required pressure readings shall be taken as specified in 3.8.1.

4.6.8.2 Temperature conditioning. Each assembled initiator shall be conditioned for not less than 4 hours nor greater than 24 hours. When reconditioning initiators at -60°F, all condensation shall be removed from the exterior surfaces before returning the initiator to the conditioning chamber.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 9252247 and 9252248. Level B shall be in accordance with Drawings 9252247 and 9252249.

5.1.1 Lot numbers. Lot numbers shall be in accordance with MIL-STD-1168.

6. NOTES

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(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and revision letter of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. Whether a first article sample is required (see 3.1, 4.4, and 6.4)

d. Test facility for first article tests (see 4.4)

e. Production test activity and lot size (see 4.1 and 4.5.1.2)

f. Directions for shipping radiographs of entire production lot along with ballistic sample to the activity conducting production lot acceptance tests (see 4.6.6)

g. Level of packing and marking (see 5.1)

h. That the safety precaution requirements of the ``Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of ``Ammunition and Explosives Ashore," OP 5 are applicable.

6.3 Hazard notice. The initiator described herein and certain of its components are flammable or explosive and present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these properties and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling, and electrically induced incidents.

6.4 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the

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pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions.

6.5.1 Ignition delay. Ignition delay time is the time measured between the firing pin striking the ignition device and the beginning of a continuous thrust rise on a thrust-time record.

6.5.2 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.6 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.7 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:

Preparing activity
Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0E38)

Review activity:
Air Force - 70

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